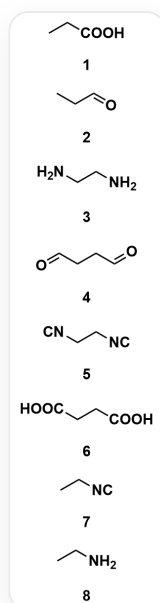


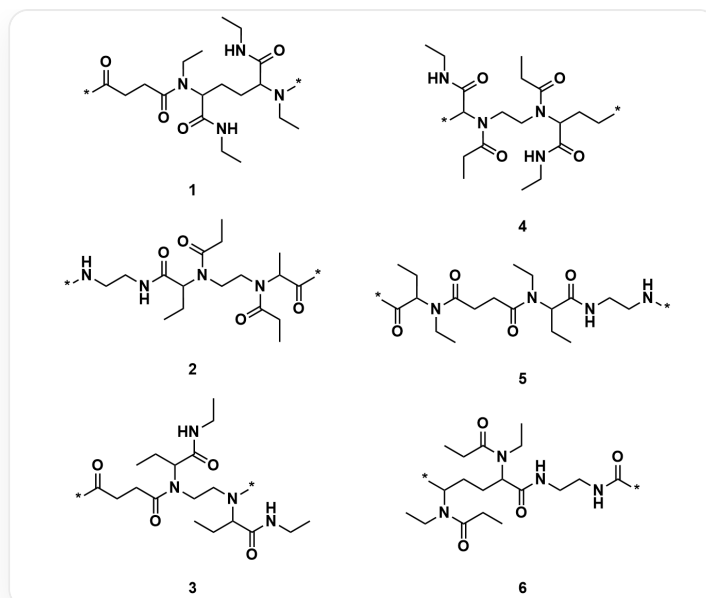
Question

The following is a series of monomer molecules A_i that can undergo polymerization reactions with each other:



The monomer molecules are as follows: 1. CCC(O)=O 2. CCC=O 3. NCCN 4. O=CCCC=O 5. [C-]#[N+]CC[N+]#[C-] 6. O=C(CCC(O)=O)O 7. CC[N+]#[C-] 8. NCC

The following is a series of repeating units P_j of polymer, which are obtained from the above monomer molecules (in the figure, `\*` indicates the connection site of the repeating unit fragment in the polymer):



The repeating units of the polymer are as follows: 1. O=C([*])CCC(N(CC)C(CCC(N(CC)[*])C(NCC)=O)C(NCC)=O)=O 2. [*]NCCNC(C(CC)N(C(CC)=O)CCN(C(C([*])=O)C)C(CC)=O)=O 3.

O=C([*])CCC(N(CCN(C(CC)C(NCC)=O)[*])C(C(NCC)=O)CC)=O 4.

O=C(NCC)C([*])N(C(CC)=O)CCN(C(C(NCC)=O)CC[*])C(CC)=O 5.

O=C([*])C(CC)N(C(CCC(N(CC)C(C(NCCN[*])=O)CC)=O)=O)CC 6.

CCN(C(CC)=O)C(CCC(N(CC)C(CC)=O)C(NCCNC([*])=O)=O)[*], where [*] indicates the connection site of the repeating unit in the polymer

For each polymer, calculate the value of z_j :

$$z_j = \prod_{i \in p_j} A_i \times \prod_{i \notin p_j} \frac{1}{A_i}$$

Where:

- p_j : The set of monomers present in the repeating unit P_j
- A_i : The serial number of monomer A_i

Finally, calculate the sum of z_j :

$$S = \sum_j z_j$$

Select the correct option, requiring the final result to retain 3 significant figures, and select the option with an error within 1% of the result; otherwise, select option **A**: All other options are incorrect.

A. All other options are incorrect

B. 52.9

C. 44.7

D. 10.1

E. 5.31

F. 2.18

G. 0.454

H. 55.8

I. 70.5

J. 91.5

K. 150

Answer

Correct Answer: **B**

Detailed Explanation

Observing the structure of the polymers, we find that each complex polymer repeating unit is generated through its corresponding four different monomers via the Ugi multi-component polymerization reaction.

CHECKPOINT

2 PTS

Each polymer is generated from four monomer molecules through Ugi multi-component polymerization

Polymer Repeating Unit	Corresponding Monomer Set (p_j)	Monomer Number
P_1	A_4, A_6, A_7, A_8	{4, 6, 7, 8}
P_2	A_1, A_2, A_3, A_5	{1, 2, 3, 5}
P_3	A_2, A_3, A_6, A_7	{2, 3, 6, 7}
P_4	A_1, A_3, A_4, A_7	{1, 3, 4, 7}
P_5	A_2, A_5, A_6, A_8	{2, 5, 6, 8}
P_6	A_1, A_4, A_5, A_8	{1, 4, 5, 8}

CHECKPOINT

1 PTS

The corresponding monomer numbers for P_1 are 4, 6, 7, 8

CHECKPOINT

1 PTS

The corresponding monomer numbers for P_2 are 1, 2, 3, 5

CHECKPOINT

1 PTS

The corresponding monomer numbers for P_3 are 2, 3, 6, 7

CHECKPOINT

1 PTS

The corresponding monomer numbers for P_4 are 1, 3, 4, 7

CHECKPOINT

1 PTS

The corresponding monomer numbers for P_5 are 2, 5, 6, 8

CHECKPOINT

1 PTS

The corresponding monomer numbers for P_6 are 1, 4, 5, 8

We will use the following formula to calculate the value of each z_j . In the formula, A_i represents the monomer number, p_j is the set of monomer numbers that constitute the polymer P_j , and we assume that each monomer appears only once in the reaction (i.e., $n_{i,j} = 1$).

$$z_j = \frac{\prod_{i \in p_j} A_i}{\prod_{i \notin p_j} A_i}$$

$$z_1 = \frac{4 \times 6 \times 7 \times 8}{1 \times 2 \times 3 \times 5} = \frac{1344}{30} \approx 44.8000$$

CHECKPOINT

0.5 PTS

$$z_1 = 44.8000$$

$$z_2 = \frac{1 \times 2 \times 3 \times 5}{4 \times 6 \times 7 \times 8} = \frac{30}{1344} \approx 0.0223$$

CHECKPOINT

0.5 PTS

$$z_2 = 0.0223$$

$$z_3 = \frac{2 \times 3 \times 6 \times 7}{1 \times 4 \times 5 \times 8} = \frac{252}{160} = 1.5750$$

CHECKPOINT

0.5 PTS

$$z_3 = 1.5750$$

$$z_4 = \frac{1 \times 3 \times 4 \times 7}{2 \times 5 \times 6 \times 8} = \frac{84}{480} = 0.1750$$

CHECKPOINT

0.5 PTS

$$z_4 = 0.1750$$

$$z_5 = \frac{2 \times 5 \times 6 \times 8}{1 \times 3 \times 4 \times 7} = \frac{480}{84} \approx 5.7143$$

CHECKPOINT

0.5 PTS

$$z_5 = 5.7143$$

$$z_6 = \frac{1 \times 4 \times 5 \times 8}{2 \times 3 \times 6 \times 7} = \frac{160}{252} \approx 0.6349$$

CHECKPOINT

0.5 PTS

$$z_6 = 0.6349$$

Finally, calculate:

$$S = \sum_{j=1}^6 z_j = z_1 + z_2 + z_3 + z_4 + z_5 + z_6$$

$$S \approx 44.8000 + 0.0223 + 1.5750 + 0.1750 + 5.7143 + 0.6349$$

$$S \approx 52.9$$

CHECKPOINT

0.5 PTS

$$S \approx 52.9$$